

AI ASSISTED CODING

Assignment-6.1

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Batch:14

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#Task-1:

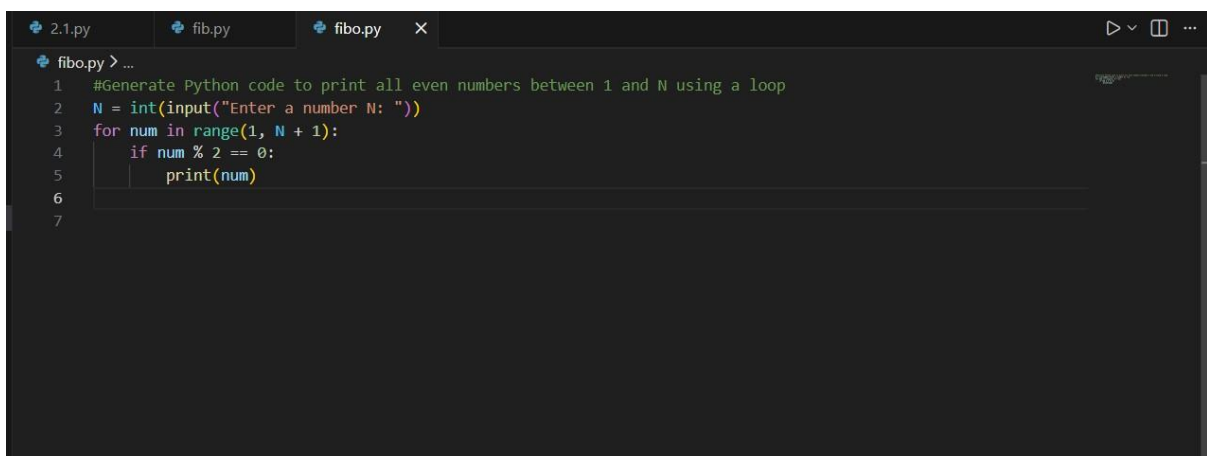
Task Description #1 (AI-Based Code Completion for Loops)

Task: Use an AI code completion tool to generate a loopbased program.

Prompt:

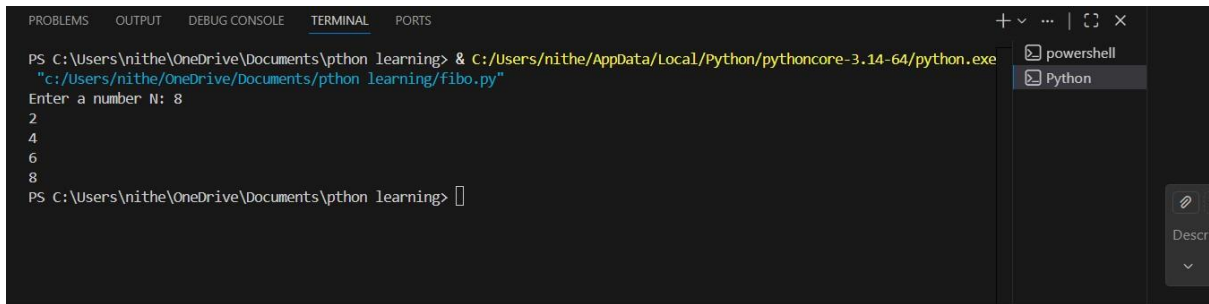
#Generate Python code to print all even numbers between 1 and N using a loop

Code:

A screenshot of a code editor with a dark theme. The editor has three tabs at the top: '2.1.py', 'fib.py', and 'fib.py' with a close button. The active tab is 'fib.py'. The code in the editor is as follows:

```
1 #Generate Python code to print all even numbers between 1 and N using a loop
2 N = int(input("Enter a number N: "))
3 for num in range(1, N + 1):
4     if num % 2 == 0:
5         print(num)
6
7
```

Output:



```
PS C:\Users\nithe\OneDrive\Documents\pthon learning> C:/Users/nithe/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/nithe/OneDrive/Documents/pthon learning/fibo.py"
Enter a number N: 8
2
4
6
8
PS C:\Users\nithe\OneDrive\Documents\pthon learning>
```

Justification:

The program uses a **for loop** to traverse numbers from 1 to N.

An **if condition** checks whether each number is even using the modulo operator.

Only numbers satisfying the condition are printed, ensuring correct identification of even numbers.

#Task-2

Task Description #2 (AI-Based Code Completion for Loop with Conditionals)

Task: Use an AI code completion tool to combine loops and conditionals.

Prompt:

#Generate a python code to count how many even and odd numbers are there in a List

Code:

```
3 #Generate a python code to count how many even and odd numbers are there in a List
4 numbers = [10, 21, 32, 43, 54, 65, 76, 87, 98]
5 even_count = 0
6 odd_count = 0
7 for number in numbers:
8     if number % 2 == 0:
9         even_count += 1
10    else:
11        odd_count += 1
12 print("Even numbers count:", even_count)
13 print("Odd numbers count:", odd_count)
14
15
16
```

Output:

```
PS C:\Users\nithe\OneDrive\Documents\python learning> & C:/Users/nithe/AppData/Local/Python/pythoncore-3.14-64/python.exe
"c:/Users/nithe/OneDrive/Documents/python learning/fibo.py"
Even numbers count: 5
Odd numbers count: 4
PS C:\Users\nithe\OneDrive\Documents\python learning> █
```

Justification:

The program uses a **for loop** to iterate through each number in the list.

An **if-else conditional** checks whether the current number is divisible by 2 to determine even or odd.

Separate counters are updated accordingly, and the final counts are displayed as output.

#Task-3:

Task Description #3 (AI-Based Code Completion for Class Attributes Validation)

Task: Use an AI tool to complete a Python class that validates user

Prompt:

#Generate a Python class User that validates age and email using conditional statements Code:

```
1 #Generate a Python class User that validates age and email using conditional statements
2 class User:
3     def __init__(self, age, email):
4         self.age = age
5         self.email = email
6
7     def validate_age(self):
8         if isinstance(self.age, int) and self.age >= 18:
9             return "Valid age"
10        else:
11            return "Invalid age"
12
13    def validate_email(self):
14        if "@" in self.email and "." in self.email:
15            return "Valid email"
16        else:
17            return "Invalid email"
18
19 # Valid input
20 user1 = User(25, "user@example.com")
21 print(user1.validate_age()) # Valid age
22 print(user1.validate_email()) # Valid email
23
24 # Invalid input
25 user2 = User(15, "userexample.com")
26 print(user2.validate_age()) # Invalid age
27 print(user2.validate_email()) # Invalid email
28
29
```

Output:

```
PS C:\Users\nithe\OneDrive\Documents\python learning> & C:/Users/nithe/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/nithe/OneDrive/Documents/python learning/fib.py"
Valid age
Valid email
Invalid age
Invalid email
PS C:\Users\nithe\OneDrive\Documents\python learning> 
```

Justification:

The AI-generated User class uses **conditional statements** to validate age and email attributes based on predefined rules.

Separate validation methods ensure **proper condition handling** for both valid and invalid inputs.

Test cases confirm that the class correctly distinguishes between acceptable and unacceptable user data.

Task-4:

Task Description #4 (AI-Based Code Completion for Classes)

Task: Use an AI code completion tool to generate a Python class for managing student details.

Prompt:

#“Generate a Python class Student with attributes (name, roll number,marks) and methods to calculate total and average marks.”

Code:

```
2
3  #Generate a Python class Student with attributes (name, roll number,marks) and methods to calculate total an
4  class Student:
5      def __init__(self, name, roll_number, marks):
6          self.name = name
7          self.roll_number = roll_number
8          self.marks = marks # marks should be a list of integers
9
10     def calculate_total(self):
11         return sum(self.marks)
12
13     def calculate_average(self):
14         if len(self.marks) == 0:
15             return 0
16         return self.calculate_total() / len(self.marks)
17 # Example usage
18 student1 = Student("Alice", 101, [85, 90, 78, 92])
19 print("Total Marks:", student1.calculate_total()) # Total Marks: 345
20 print("Average Marks:", student1.calculate_average())
21 print("Roll number:", student1.roll_number) # Roll number: 101
22 student2 = Student("Bob", 102, [70, 75, 80])
23 print("Total Marks:", student2.calculate_total()) # Total Marks: 225
24 print("Average Marks:", student2.calculate_average()) # Average Marks: 75.0
25 print("Roll number:", student2.roll_number) # Roll number: 102
26 student2
27
28
29
30
```

Output:

```
PS C:\Users\nithe\OneDrive\Documents\pthon learning> & C:/Users/nithe/AppData/Local/Python/pythoncore-3.14-64/python.exe
"c:/Users/nithe/OneDrive/Documents/pthon learning/fibo.py"
Total Marks: 345
Average Marks: 86.25
Roll number: 101
Total Marks: 225
Average Marks: 75.0
Roll number: 102
PS C:\Users\nithe\OneDrive\Documents\pthon learning> 
```

Justification:

The AI-generated Student class correctly defines attributes and methods, ensuring a **complete and wellstructured class design**.

Methods for calculating total and average marks verify the **correctness of logic and data handling** within the class.

Task-5:

Task Description 5 (AI-Assisted Code Completion Review)

Task: Use an AI tool to generate a complete Python program using classes, loops, and conditionals together.

Prompt:

#Generate a Python program for a simple bank account system using class, loops, and conditional statements

Code:

```

1  #Generate a Python program for a simple bank account system using class,loops, and conditional statements
2  class BankAccount:
3      def __init__(self, account_holder, balance=0):
4          self.account_holder = account_holder
5          self.balance = balance
6
7      def deposit(self, amount):
8          if amount > 0:
9              self.balance += amount
10             print(f"Deposited: {amount}. New balance: {self.balance}")
11         else:
12             print("Deposit amount must be positive.")
13
14     def withdraw(self, amount):
15         if amount > 0:
16             if amount <= self.balance:
17                 self.balance -= amount
18                 print(f"Withdrew: {amount}. New balance: {self.balance}")
19             else:
20                 print("Insufficient funds.")
21         else:
22             print("Withdrawal amount must be positive.")
23
24     def display_balance(self):
25         print(f"Account holder: {self.account_holder}, Balance: {self.balance}")
26
27 # Example usage
28 account = BankAccount("John Doe", 1000)
29 account.display_balance()
30 while True:
31     print("\nOptions: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit")
32     choice = input("Enter your choice (1-4): ")
33
34     if choice == '1':
35         amount = float(input("Enter amount to deposit: "))
36         account.deposit(amount)
37     elif choice == '2':
38         amount = float(input("Enter amount to withdraw: "))

```

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```

26 # Example usage
27 account = BankAccount("John Doe", 1000)
28 account.display_balance()
29 while True:
30     print("\nOptions: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit")
31     choice = input("Enter your choice (1-4): ")
32
33     if choice == '1':
34         amount = float(input("Enter amount to deposit: "))
35         account.deposit(amount)
36     elif choice == '2':
37         amount = float(input("Enter amount to withdraw: "))
38         account.withdraw(amount)
39     elif choice == '3':
40         account.display_balance()
41     elif choice == '4':
42         print("Exiting the program.")
43         break
44     else:
45         print("Invalid choice. Please try again.")
46
47

```

Output:


```
PS C:\Users\nithe\OneDrive\Documents\python learning> & C:/Users/nithe/AppData/Local/Python/pythoncore-3.14-64/python.exe  
"c:/Users/nithe/OneDrive/Documents/python learning/2.1.py"  
Account holder: John Doe, Balance: 1000  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
Enter your choice (1-4): 3  
Account holder: John Doe, Balance: 1000  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
Account holder: John Doe, Balance: 1000  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
Enter your choice (1-4): 3  
Account holder: John Doe, Balance: 1000  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
Enter your choice (1-4): 3  
Account holder: John Doe, Balance: 1000  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit  
Enter your choice (1-4): 3  
Account holder: John Doe, Balance: 1000  
  
Options: 1. Deposit 2. Withdraw 3. Display Balance 4. Exit
```

Justification:

The AI-generated bank account program integrates **classes**, **loops**, and **conditional statements** to model real-world banking operations.

Loops enable repeated user interactions, while conditionals ensure **valid decision-making** for deposits, withdrawals, and balance checks.

